

Xiaoyu Wang

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[Personal Website](#) | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

EDUCATION

Ph.D. in Biological Systems Engineering

Jan 2023 – Present

University of Wisconsin–Madison, Madison, WI

Minor: Computer Sciences

Relevant Coursework: CS 533 (Image Processing); CS 760 (Machine Learning); CS 759 (High Performance Computing for Applications in Engineering)

Doctoral Studies in Computing and Information Sciences

Aug 2022 – 2023

Rochester Institute of Technology, Rochester, NY

Golisano College of Computing and Information Sciences

Relevant Coursework: CSCI 820 (Quantitative Foundations); CSCI 865 (Deep Learning)

B.S. in Computer Science and Technology

2017 – 2021

Xi'an Jiaotong University, Xi'an, China

EXPERIENCE

Graduate Research Assistant

Jan 2023 – Present

University of Wisconsin–Madison, Madison, WI

- Developed and deployed a full-stack cyber-agricultural web platform using Vue 3, MapLibre GL JS, FastAPI, and Vite to publish interactive county-level crop-yield predictions.
- Built geospatial visualization and backend API workflows that deliver in-season county estimates ahead of USDA NASS official releases for researchers, policymakers, analysts, and agricultural stakeholders.
- Structured a frontend/backend monorepo and production workflow with Vite proxying, Gunicorn serving, and deployment automation; launched the live [Web Tool](#) and released the [source code](#).

Research Intern

Apr 2021 – Apr 2022

Microsoft Research Asia, Media Computing Group, Beijing, China

- Designed a multi-modal multi-correlation learning framework for audio-visual speech separation that explicitly modeled speaker-identity and phonetic correlations using contrastive and adversarial training.
- Integrated a U-Net separation backbone with wav2vec 2.0, lip-reading, and ResNet-18 speaker-identification models in PyTorch; trained and evaluated models on four NVIDIA V100 GPUs.
- Improved LRS3 SDR from 8.46 to 9.98, PESQ from 2.27 to 2.584, and STOI from 0.843 to 0.861 without increasing inference complexity; published a [first-author paper at Interspeech 2022](#).

Research Intern

Sep 2020 – Mar 2021

SenseTime, Autonomous Driving Group

- Contributed to [OpenCalib](#), an open-source multi-sensor calibration toolbox for autonomous driving, by implementing calibration workflows among LiDAR, radar, and camera sensors.
- Trained a deep learning model to determine whether multi-sensor calibration results were accurate, automating calibration-quality assessment.

Remote Research Intern

Jun 2020 – Sep 2020

Nanyang Technological University

- Proposed a data-free targeted universal adversarial perturbation method with a custom loss that balanced model disruption and target-label objectives without access to training data.
- Extended the attack to federated-learning settings and published the work as a [first-author paper at SciSec 2021](#).

Research Intern INNNO

Jun 2019 – Feb 2020

- Built a remote-sensing change-detection pipeline for drone imagery, including image alignment, bi-temporal image preprocessing, and semantic-segmentation mask prediction.
- Created synthetic change pairs to address limited labeled data and trained a Python-based model to detect changes in buildings, waterways, and vegetation; released the [project code](#).

Research Assistant

Jan 2019 – Sep 2019

College of Artificial Intelligence, Xi'an Jiaotong University

- Developed Distributed-GAN, enabling multiple users to train generative models locally on private data without sharing raw datasets while improving generated-sample diversity.
- Published the work as a [first-author arXiv preprint](#).

Research Assistant

Oct 2017 – 2018

Xi'an Jiaotong University

Conducted research on computer security and deep learning.

PUBLICATIONS

Journal Articles

Assimilation of LAI and FAPAR within DSSAT Guided by Leaf Growth Dynamics to Enhance Maize Yield Phenotyping 2026

Zezhong Tian, Jiahao Fan, Tong Yu, **Xiaoyu Wang**, Mingda Wu, Natalia de Leon, Shawn M. Kaeppler, Zhou Zhang
Computers and Electronics in Agriculture, vol. 250, article 111946, 2026.

Learning County from Pixels: Corn Yield Prediction with Attention-Weighted Multiple Instance Learning 2025

Xiaoyu Wang, Yuchi Ma, Yijia Xu, Qunying Huang, Zhengwei Yang, Zhou Zhang
International Journal of Remote Sensing, vol. 46, no. 7, pp. 2815–2845, 2025.

Book Chapter

Satellite-Based Mapping of Crop Yield 2026

Yuchi Ma, Shuo Chen, **Xiaoyu Wang**, Zhou Zhang, Zhengwei Yang, Qunying Huang, Xinghua Cheng, Gengchen Mai
In Agricultural Applications of Earth Observation, ch. 7, pp. 193–230, Elsevier, 2026.

Conference Papers

Knowledge-Guided Machine Learning Model with Soil Moisture for Corn Yield Prediction under Drought Conditions 2025

Zhengwei Yang, **Xiaoyu Wang**, Jingyi Huang, Zhou Zhang
Accepted to IEEE International Geoscience and Remote Sensing Symposium (IGARSS), 2025.

County-Level Crop Yield Prediction Using SMAP-Derived Data Products and Deep Learning Model 2024

Zhengwei Yang, **Xiaoyu Wang**, Jingyi Huang, Zhou Zhang
IEEE International Geoscience and Remote Sensing Symposium (IGARSS), pp. 3222–3225, 2024.

Multi-Modal Multi-Correlation Learning for Audio-Visual Speech Separation 2022

Xiaoyu Wang, Xiangyu Kong, Xiulian Peng, Yan Lu
Interspeech, 2022.

A Data-Free Approach for Targeted Universal Adversarial Perturbation 2021

Xiaoyu Wang, Tao Bai, Jun Zhao
Science of Cyber Security (SciSec), pp. 126–138, 2021.

Preprint

Knowledge-Guided Machine Learning Model with Soil Moisture for Corn Yield Prediction under Drought Conditions 2025

Xiaoyu Wang, Yijia Xu, Jingyi Huang, Zhengwei Yang, Zhou Zhang
arXiv:2503.16328, 2025.

AWARDS & HONORS

Stars of Tomorrow Excellent Intern Award, Microsoft Research Asia, 2021.

BSE Travel Award, University of Wisconsin–Madison, 2025. (\$1,000)

Third-Class Award, Xi'an Jiaotong University, 2017. (Top 20% by GPA)

PROFESSIONAL SERVICE

Journal Reviewer

International Journal of Applied Earth Observation and Geoinformation (JAG)

TECHNICAL SKILLS

Languages: Python, C++, Shell, MATLAB, CUDA C

Machine Learning & PyTorch, TensorFlow, TensorRT, CUDA, cuDNN, cuBLAS

GPU:

Web & Deployment: FastAPI, Vue 3, MapLibre GL JS, Vite, Gunicorn, Docker

GIS & Remote Sensing: Google Earth Engine, GDAL, QGIS, ArcGIS

Vision, Audio & OpenCV, librosa, Asteroid, Kaldi, FFmpeg, PCL, ROS

Robotics:

Development: Linux, Git, CMake